CPS 843 (CP 8307) Problem Set 2

Due Monday, October 20th, @ 11:59pm
(5 assignments in total, final grade is evaluated based on the best 4 out of 5)

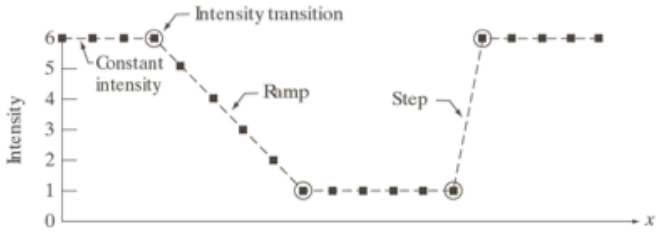
Purpose

- Familiar with programming for image processing
- Understand the principle of edge detection and image enhancement

Part 1:

Question 1. Write down the masks for Robert, Prewitt, and Sobel edge detectors. Then, take or download an image and perform edge detection using these three edge detectors, respectively. Show your results and make a reasonable analysis of your results. (4 pts)

Question 2. Given a 1D image $f(x)$, write down the 1st order and 2nd order derivatives of the function $f(x)$. Then, manually compute the 1st order and 2nd order derivatives of the following image (use zero-padding if necessary). (4 pts)



Question 3. Write down the image sharpening process based on unsharp masking and high-boost filtering. Take or download an image, and sharpen the image based on the high-boost filtering (choose k=1 and k=5). Show your results and make a reasonable analysis. (4 pts)

Question 4. Take or download an image and convert it to a grayscale image. Add Gaussian noise to the image, then use the average filter and the Gaussian filter to remove the noise. Choose two different noise levels and select proper filter sizes to achieve reasonable performance. Show and analyze your results. (4 pts)

Question 5. (1) Write down the equations of the 1st-order derivative $\frac{\partial f(x,y)}{\partial x}$, $\frac{\partial f(x,y)}{\partial y}$ and the Laplacian of a 2D image $f(x,y)$. (2) Compute the 1st derivative of the following 3-bit image directly using the equations of (1). (3) Compute the Laplacian of the following 3-bit image using the given Laplacian mask. (4) normalize the Laplacian result. (Use zero-padding if needed, do the problem manually and show necessary steps) (4 pts)

0	1	0
1	−4	1
0	1	0

0	2	5	7
2	5	7	3
5	6	3	1
5	2	1	0

Part 2: (4 pts)

Software:
Matlab example for low-light image enhancement
https://www.mathworks.com/help/images/low-light-image-enhancement.html

Work to do:
- Carefully read and understand the basic ideas presented in the paper "Fast Efficient Algorithm for Enhancement of Low Lighting Video," available on D2L under the "Assignments" folder.
- Download or capture two low-light images and resize them if necessary to ensure they are suitable for processing.
- Follow the steps outlined in the example to enhance your two images using different enhancement approaches.

Report:
- A brief technical description of the image enhancement algorithm (in about a half page).
- A description of each step, the source code, and the corresponding results.
- A brief analysis of your results.

Part 3: (1 pt)

What is the tentative topic of your final project?

Requirements

- The assignment is due on Monday, October 20th @ 11:59pm. You have 24 hours of grace time without penalty. Submissions after the grace period will not be accepted.
- Submit all your work in a single PDF file through D2L (multiple submissions are allowed, but only the last submission will be considered for evaluation).
- Please resize all images properly in line with the text of your report.
- Submit the source code in line with the report of each part.
- You can directly use any available functions of Matlab for your work. Other programming languages and packages are also allowed.
- Complete the report by yourself. We will use Turnitin® for similarity check.
- Suggested report format: IEEE double-column format. The Word and LaTeX template can be found via the link below (please note that any other formats are also allowed)
  http://www.ieee.org/conferences_events/conferences/publishing/templates.html

TA for HW#2:

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You may contact the TA directly if you have any questions about this assignment.